Fiscal Year:	FY 2012	**Task Last Updated:**	FY 08/02/2011
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
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City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	10/01/2009	**End Date:**	09/30/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	4
No. of PhD Candidates:	2	**No. of Master' Degrees:**	4
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
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Flight Program:	ISS		
Flight Assignment:	NOTE: Project to be extended to 9/30/2014 per D. Stocker/GRC (Ed., 7/20/2011)		
Key Personnel Changes/Previous PI:	Not applicable.		
COI Name (Institution):	Chao, Beei-Huan (University of Hawaii at Manoa) Sunderland, Peter (University of Maryland) Urban, David (NASA Glenn Research Center)		
Grant/Contract No.:	NNX09AO53A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the PI, the three Co-I's, and staff at NASA Glenn Research Center. The division of labor has PBS and DLU performing microgravity experiments at the NASA Glenn Research Center, PBS directing numerical simulations at University of Maryland, BHC performing analytical work and computations at University of Hawaii, and RLA performing normal gravity experiments and computations at Washington University.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	• The research team published three peer-reviewed journal articles during this period: Skeen SA, Yablonsky G, Axelbaum RL. "Characteristics of non-premixed oxygen-enhanced combustion: II. Flame structure effects on soot precursor kinetics resulting in soot-free flames." Combustion and Flame, 2010 Sep;157(9):1745-52. http://dx.doi.org/ Lecoustre VR, Sunderland PB, Chao BH, Axelbaum RL. "Numerical investigation of spherical diffusion flames at their sooting limits." Combustion and Flame. In Press, Corrected Proof, Available online 29 June 2011. http://dx.doi.org/ Wang Q, Chao BH. "Kinetic and radiative extinctions of spherical burner-stabilized diffusion flames." Combustion and Flame 2011 Aug;158(8):1532-41. http://dx.doi.org/ • Work at Washington University continued to pursue normal gravity experiments of soot formation in diffusion flames, numerical modeling, and chemical kinetics. • Work at Univ. of Maryland continued to pursue numerical models of spherical diffusion flames, including soot formation and extinction. • Work at Univ. of Hawaii continued to pursue analytical and computational studies of soot formation, radiation, and extinction in diffusion flames. • The flight experiment was supported in various ways. For example, the research team responded to issues raised during the Requirements Definition Review (RDR) for this project. The research team also provided guidance on the ACME operational requirements.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Other Journals or Periodicals	Lecoustre VR, Sunderland PB, Chao BH, Axelbaum RL. "Numerical investigation of spherical diffusion flames at their sooting limits." Combustion and Flame. 2012 Jan;159(1):194-9. https://doi.org/10.1016/j.combustflame.2011.05.022 , Jan-2012
Articles in Peer-reviewed Journals	Wang Q, Chao BH. "Kinetic and radiative extinctions of spherical burner-stabilized diffusion flames." Combustion and Flame. 2011 Aug;158(8):1532-41. https://doi.org/10.1016/j.combustflame.2010.12.007 , Aug-2011
Articles in Peer-reviewed Journals	Skeen SA, Yablonsky G, Axelbaum RL. "Characteristics of non-premixed oxygen-enhanced combustion: II. Flame structure effects on soot precursor kinetics resulting in soot-free flames." Combustion and Flame, 2010 Sep;157(9):1745-52. http://dx.doi.org/10.1016/j.combustflame.2010.04.013 , Sep-2010
Papers from Meeting Proceedings	Lecoustre VR, Sunderland PB, Chao BH, Axelbaum RL. "Characterization of microgravity spherical diffusion flame sooting limits." 7th U.S. National Combustion Meeting, Atlanta, GA, March 20-23, 2011. 7th U.S. National Combustion Meeting, Atlanta, GA, March 20-23, 2011. 6 p. , Mar-2011
Papers from Meeting Proceedings	Lecoustre VR, Sunderland PB, Chao BH, Axelbaum RL. "Numerical study of laminar spherical hydrogen diffusion flames at low flow rates." 7th U.S. National Combustion Meeting, Atlanta, GA, March 20-23, 2011. 7th U.S. National Combustion Meeting, Atlanta, GA, March 20-23, 2011. 6 p. , Mar-2011